Native arithmetic replay: subgroup-preserving paired workload

Replay snapshot 2026-10-25T04:50:58.997Z: 4600/4600 rho and 4600/4600 BSGS cases have passed an independent public point equation and matched the original answer plus every non-clock native statistic. The complete selected component has 46 vertices and 100 paired problems per vertex. Missing cases remain hatched, not zero. Partial coverage must not be interpreted as a matched comparison of means.

These separately frozen binaries count source-level modular primitive calls in thread-local counters. The native algorithm receives only the public P/Q problem and public walk seed; baseline answers are read only after the native API returns. Original binaries, input identities, case timings and checkpoints remain unchanged. Replay wall/process CPU timers belong to the instrumented replay, not the baseline.

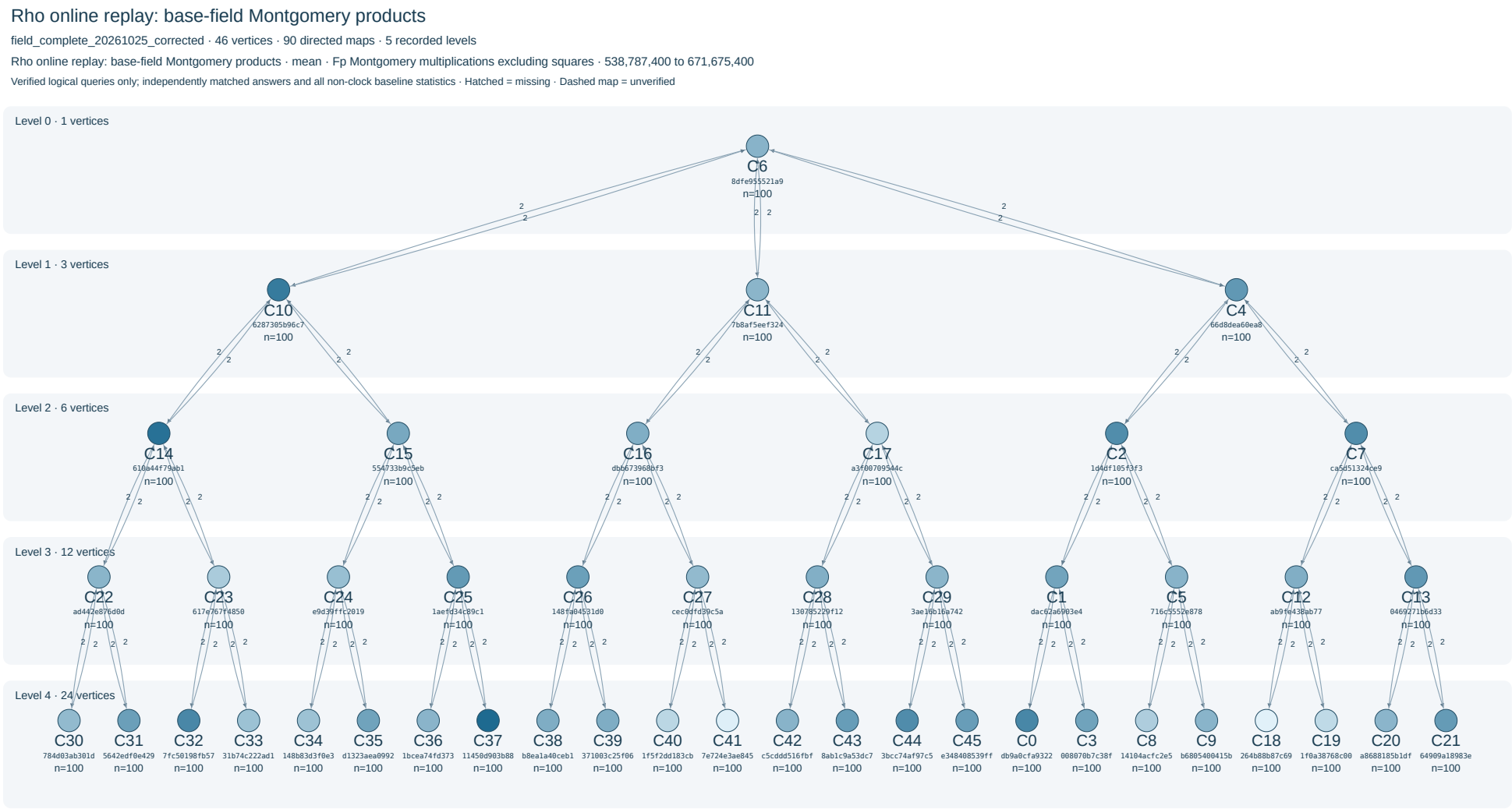
Fp, scalar-mod-r and other-modulus counters separately record add, subtract, plain multiplication, Montgomery multiplication, Montgomery squaring, reduction, inversion, Euclidean division and complement calls. Montgomery multiplication and squaring are disjoint counters. Their per-case sum is a derived total. Montgomery reductions overlap products/conversions; do not sum these columns as independent work. Inversions use extended Euclid, unlike the reference-map inverse exponentiation. These are not CPU instruction counts; the host exposes no hardware PMU for the attempted events.

Per-curve, per-generator and per-problem tables retain min/mean/max and observation counts for every primitive and replay timer. Setup and destruction are separately retained for every attempt. Counted rho runs use one thread, allowing complete thread-local capture. Repeated/interrupted work stays in the attempt ledger rather than being silently discarded.

Sources: FIELD_PROTOCOL.md, FIELD_FROZEN.json, source/build_field_backend.mjs, the readable field_backend tree, source/field_worker.cpp and source/field_aggregate.mjs. Original timing and full-path transport reports keep their own boundaries. This concurrent L0 replay makes no fastest, isolated timing or intrinsic-hardness claim. Full random-corpus IC recovery remains unresolved; see IC_RECOVERY_GAP.md.

Rho online replay: base-field Montgomery products

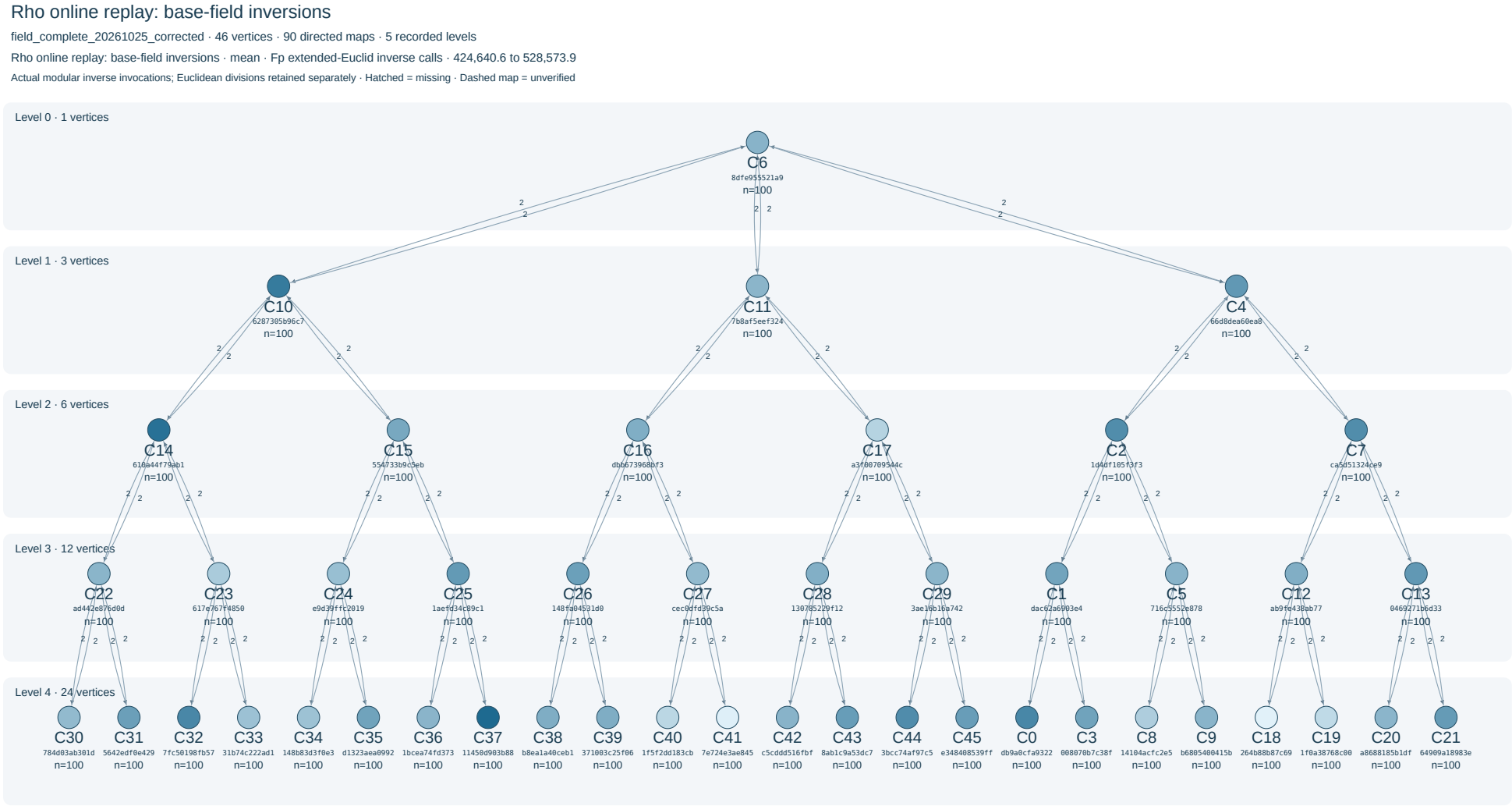
Verified logical queries only; independently matched answers and all non-clock baseline statistics. Unit: Fp Montgomery multiplications excluding squares. Exact statistics and curve identities: rho_fp_products/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Rho online replay: base-field inversions

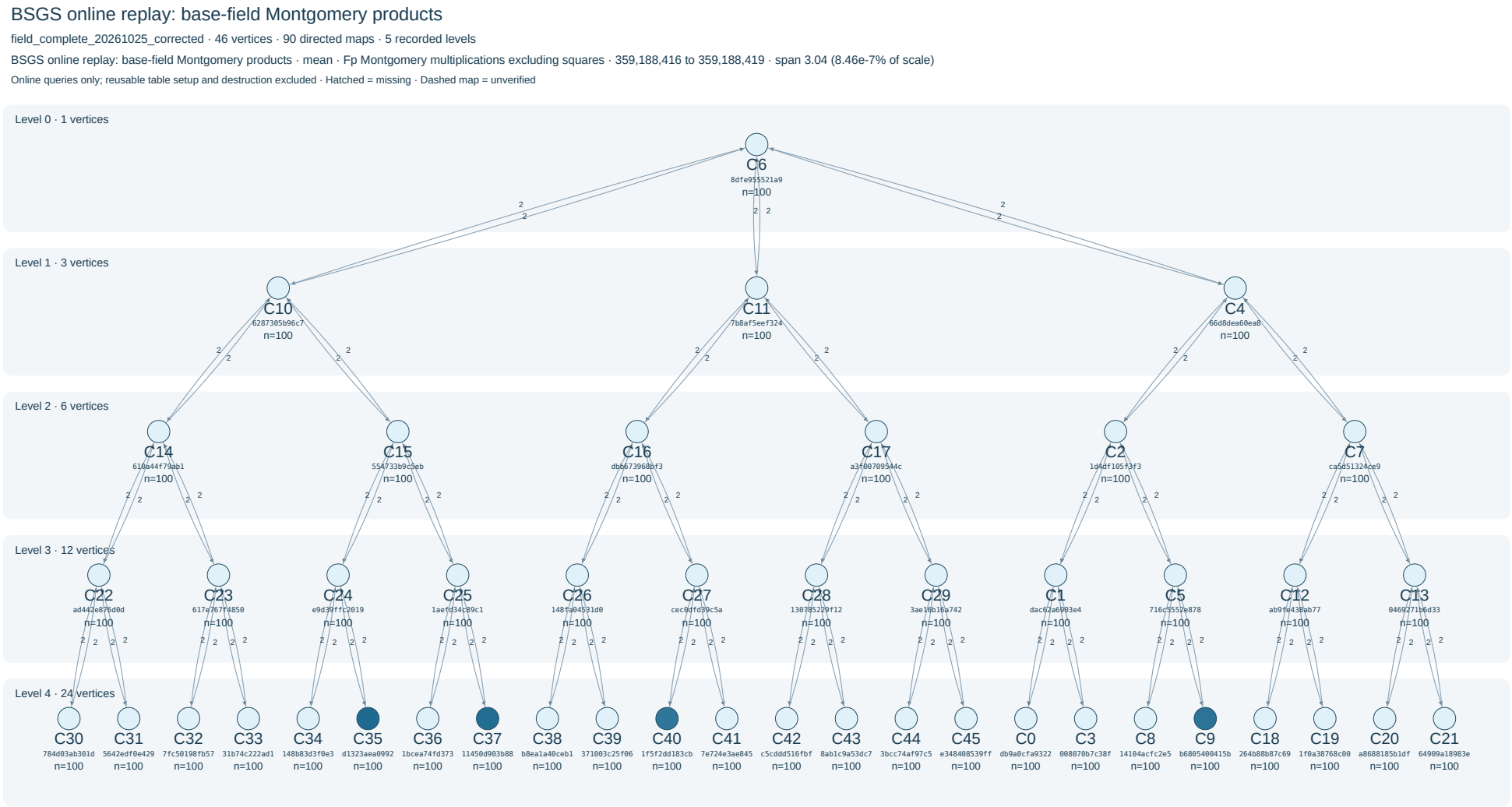
Actual modular inverse invocations; Euclidean divisions retained separately. Unit: Fp extended-Euclid inverse calls. Exact statistics and curve identities: rho_fp_inversions/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

BSGS online replay: base-field Montgomery products

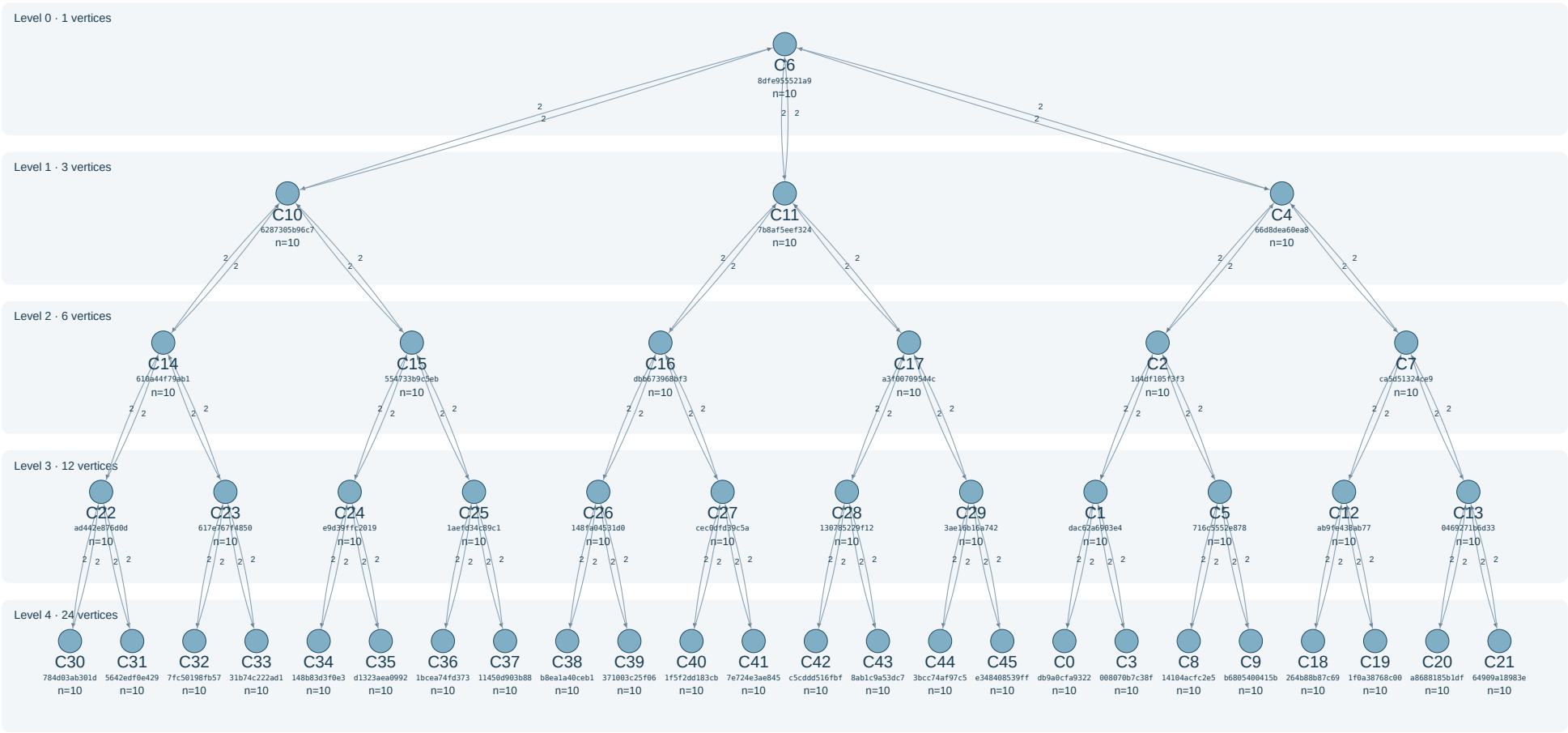
Online queries only; reusable table setup and destruction excluded. Unit: Fp Montgomery multiplications excluding squares. Exact statistics and curve identities: bsgs_fp_products/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

BSGS table setup replay: base-field Montgomery products

One observation per recorded setup attempt; all attempts retained, including rebuilds. Unit: Fp Montgomery multiplications excluding squares. Exact statistics and curve identities: bsgs_setup_products/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.